

BTICS601
B.Tech.(CSE-ICS)
VI Semester Examination, Dec.23-Jan.24
Concepts of System Security
Choice Based Credit System (CBCS)

Time: 3 Hrs.

Maximum Marks: 60

Minimum Pass Marks: 24

*Note: (1) All questions carry equal marks, out of which part 'A' and 'B' carry 3 marks and part 'C' carries 6 marks.
(2) From each question, part 'A' and 'B' are compulsory and part 'C' has internal choice.
(3) Draw neat diagram, wherever necessary.
(4) Assume suitable data wherever necessary.*

- Q.1.(A)** Define the three pillars of security known as CIA. Why are they considered fundamental in system security? 03
- (B)** What is the relationship between the terms attack, threat, vulnerability, and risk in the context of system security? 03
- (C)** Discuss the concept of system policy and its role in maintaining security within an organization. 06

OR

How does the principle of easiest penetration guide security measures in a system?

- Q.2.(A)** What is hardware vulnerability, and how can it impact the overall security of a system? 03
- (B)** Explain the concept of data vulnerability and its relevance to information security. 03
- (C)** What is Multics, and what are its fundamental characteristics in the realm of computer security? 06

OR

Explain how network vulnerabilities can be exploited by attackers?

- Q.3.(A)** What is a secure operating system, and how does it differ from a regular operating system in terms of security features? 03
- (B)** Define the concept of a trust model in the context of operating system security. 03
- (C)** Explain the concept of a mandatory protection system and its role in enforcing security policies. 06

OR

Discuss Lampson's Access Matrix and its significance in access control within an operating system.

Contd.....

- Q.4.(A)** How does UNIX address security concerns in its operating system architecture? **03**
- (B)** Define the concept of information flow in the context of security goals. How does it relate to the movement of data within a system? **03**
- (C)** What are the potential risks posed by covert channels, and how can operating systems mitigate these risks? **06**

OR

How does the information flow integrity model contribute to preventing unauthorized modifications to data?

- Q.5.(A)** What is smartphone security, and why is it crucial in the current digital landscape? **03**
- (B)** Define repackaging attacks in the context of smartphone security. **03**
- (C)** Explain the concept of whole-disk encryption in the context of smartphone security. **06**

OR

What are some emerging threats in smartphone security, and how can users stay ahead of these challenges?

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